Exploring Recurrent Architectures for EMG-to-QWERTY Decoding

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Abstract

This report is prepared for the ECE C147/C247 final project on the emg2qwerty dataset. Our project studies how recurrent sequence models decode typed characters from wrist electromyography (EMG) signals, with a primary focus on comparing vanilla RNN, LSTM, and GRU architectures under the subject 89335547.

1 Introduction

The central goal of this project is to identify and train a model that achieves lower loss and better character error rate (CER) than the base TDS model under the final project guidelines. Because of limited computing resources, all of our experiments were conducted on a single subject, user 89335547. We also kept our models relatively small, targeting fewer than 5 million parameters, both to reduce the risk of overtraining on a limited dataset and to allow faster iteration within the project timeline.

This setting naturally motivated us to explore recurrent architectures. The emg2qwerty task maps continuous wrist sEMG signals to an unsegmented character sequence, so the temporal structure of the signal is central to successful decoding. In contrast to the baseline TDS model, which is primarily convolutional, recurrent neural networks (RNNs) offer a direct mechanism for processing information over time. We therefore focused first on a vanilla RNN as a lightweight temporal baseline, then considered gated recurrent variants such as gated recurrent units (GRUs) and long short-term memory (LSTM) networks to improve optimization and temporal credit assignment on limited data. We also considered a hybrid CNN+RNN architecture as a way to combine spatial feature extraction with temporal modeling if purely recurrent models proved insufficient.

In addition to architecture choice, we considered data augmentation and input reduction strategies to improve generalization without substantially increasing model size. In particular, we explored whether reducing signal detail through lower-frequency resampling or restricting the model to subsets of available channels could encourage the models to rely on more robust patterns rather than subject-specific noise. Together, these design choices frame the main question of this report: which compact recurrent or hybrid architecture most effectively improves decoding quality over the base TDS model on the provided single-user split?

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Table 1: Official single-user split used for the main experiments

Split	Sessions	User ID	Notes
Train	16	89335547	Official training sessions from the provided config
Validation	1	89335547	Used for model selection and early comparison
Test	1	89335547	Held out for the final architecture comparison

Table 2: Model families used in the architecture comparison

Model	Encoder configuration	MLP features	Dropout
TDS baseline	4 TDS convolution blocks, kernel width 32	[384]	–
RNN	2-layer bidirectional RNN, hidden size 256	[256]	0.0
GRU	4-layer bidirectional GRU, hidden size 128	[256, 256]	0.1
Hybrid CNN+RNN	4 TDS blocks + 4-layer bidirectional GRU, hidden size 192	[192, 192]	0.1

2 Methods

2.1 Dataset and evaluation setup

This project uses the emg2qwerty dataset released by Meta Reality Labs [2]. For the main comparison, we follow the official single-user split provided by the course codebase for user 89335547. All four model families are trained and evaluated on the same 16/1/1 train/validation/test session split so that differences in performance can be attributed primarily to encoder design rather than to changes in data availability or subject identity.

We evaluate all models primarily with character error rate (CER), which is the main metric emphasized in the project handout and is well matched to the CTC decoding objective. We also track insertion, deletion, and substitution error rates to better diagnose failure modes, since different architectures may over-predict characters, miss characters entirely, or confuse similar temporal patterns in different ways. Validation CER is used for model selection, while the final held-out test session is reserved for the architecture comparison reported in the Results section.

2.2 Model families

Our architecture study compares four encoder families that share the same windowed EMG front end and CTC decoding pipeline. The provided TDSConv model serves as the course baseline and offers a strong convolutional reference point for the task. We then compare three alternatives developed around explicit temporal recurrence: a vanilla bidirectional RNN as the simplest recurrent baseline, a deeper bidirectional GRU to test whether gating improves stability and sequence modeling, and a hybrid CNN+RNN model that applies TDS-style convolutional blocks before a recurrent encoder. This comparison is designed to test whether temporal recurrence alone is sufficient to outperform the baseline or whether a mixed convolutional-recurrent encoder yields a better trade-off on the limited single-user setting.

All four models operate on the same log-spectrogram EMG representation and produce framewise character logits for greedy CTC decoding. This shared output structure lets us compare the effect of encoder choice without changing the loss, decoder, or evaluation metric between runs.

2.3 Vanilla RNN

We first picked an RNN with no modifications to fit the data. An RNN is a neural network with the ability to retain information in the short-term from past data using a feedback loop. Because of this, we believed that this model has potential to fit sEMG data well, given that sEMG data has a time dimension. Being the most simple form of RNN, we also used this model as a starting point for further improvements we anticipate needing. These improvements include something to address the fact that RNNs often struggle with vanishing gradients. Gradients sometimes vanish, or become too small and therefore negligible, when being passed through too many layers. If the input data is long enough (temporally), this could be an issue, as the RNN would slowly lose earlier data. We used a bidirectional RNN to allow the model to have a memory of both past and future data. We also used dropout to prevent overfitting. It was after inadequate empirical performance that we considered a more advanced RNN model.

2.4 RNN with GRU

After the disappointing performance of RNN, we investigated using an RNN with Gated Recurrent Units (GRU). We chose GRU over LSTM due to GRU requiring less parameters, with us moving on to LSTM in case GRU still did not perform adequately. In theory, RNN with GRU will have better performance than a vanilla RNN due to the long length of the sEMG data. We suspect part of the performance issues with a vanilla RNN is from vanishing gradients, an issue GRU does not suffer from as much.

2.5 Hybrid TDS with GRU

To begin, we noticed that the baseline model performed decently. However, it was hampered by its inability to process temporal information, the way an RNN can. From here, we thought of combining the base TDS model with the best RNN model we had at the time, which was an RNN with GRU. The idea here is that the initial convolutional layer could extract spacial features from the sEMG data, while the recurrent layer could process temporal relationships between said spacial featuers.

We started with the baseline parameters for the TDS, and the best parameters for GRU. This included the parameters for dropout (0.1) and bidirectional flow (True).

2.6 Data Augmentation

The baseline model already makes use of some data augmentation. These include Temporal Alignment Jitter, Random Band Rotation, and SpecAugment. All of these techniques introduce noise into the sEMG data, which helps with generalization. Quick ablation testing using the baseline model shows that these techniques do indeed help with accuracy. For all of our trials, we keep these baseline techniques.

New techniques that build off from the baseline include resampling the data at a lower frequency than the baseline 2000Hz and deactivating some of the channels. We considered resampling the data because this approach would have a smoothing effect across the sEMG data. In theory, resampling at a lower frequency would remove many of the outliers present in sEMG data, potentially improving model performance.

We also considered disabling a subset of the training data, but theorized that this would not improve training. We were not training on much data to begin with. Our models typically struggled with overfitting the training data, and in this situation, we should look towards strategies to either find more data or generalize instead of removing data. Empirically, this resulted in similar results but multiple times more epochs to train compared to the base case. We did not try this technique again.

2.7 Electrode Channel Selection Experiments

In addition to architectural comparisons, we evaluated how the number and spatial distribution of EMG electrode channels affect decoding performance. The default configuration uses all 16 channels available in the dataset. To study redundancy between neighboring electrodes, we trained the hybrid CNN+GRU model using several reduced channel subsets. The subsets tested were the first half of channels [0–7], the second half [8–15], an evenly spaced subset of even channels [0,2,4,6,8,10,12,14], and a smaller evenly spaced subset [0,4,8,12]. All other preprocessing, training hyperparameters, and model architecture settings were kept fixed so that differences in performance could be attributed primarily to the choice of input channels.

2.8 Training protocol and hyperparameters

Across experiments, we keep the data pipeline fixed and vary only the encoder family and its associated hyperparameters. Input windows have length 8000 with left-context/right-context padding of 1800 and 200 samples, respectively. Raw left and right EMG streams are converted to tensors and transformed into log-spectrogram features using $n\_fft = 64$ and hop length 16. During training, we

Table 3: Training configurations for the compared model families

Model	LR	Batch size	Epoch budget	Scheduler	Notes
TDS baseline	1e-3	32	40	Warmup + cosine	Original course baseline
RNN	3e-4	32	40	Cosine annealing	2-layer bidirectional recurrent encoder
GRU	1e-4	32	40	Cosine annealing	4-layer gated recurrent encoder
Hybrid CNN+RNN	3e-4	32	40	Cosine annealing	TDS front end + bidirectional GRU encoder

Table 4: Architecture comparison with validation and test metrics

Model	Val loss $\downarrow$	Val CER $\downarrow$	Val DER	Val IER	Val SER	Test loss $\downarrow$	Test CER $\downarrow$	Test DER	Test IER	Test SER
RNN	1.46	48.89	11.099	15.18	22.62	1.191	39.83	1.79	18.69	19.34
GRU	0.626	19.67	2.99	4.36	12.32	0.625	20.60	3.24	2.46	14.89
Hybrid CNN+RNN	0.531	16.22	1.68	3.17	11.36	0.511	16.90	1.73	2.49	12.69
TDS baseline	0.757	23.17	2.02	7.53	13.6	0.798	24.34	1.84	6.09	16.40

apply the augmentations defined in the codebase, including random band rotation, temporal alignment jitter, and SpecAugment. Validation and test runs use the deterministic evaluation pipeline without these stochastic augmentations.

Optimization follows the PyTorch Lightning and Hydra configuration scaffold used throughout the repository. Unless otherwise noted, runs use batch size 32, four dataloader workers, Adam optimization, and checkpoint selection based on validation CER. Each model uses the CTC greedy decoder for evaluation, which keeps decoding consistent across the convolutional, recurrent, and hybrid architectures. Exact CER and loss values are deferred to the Results section, but the main training settings for the compared models are summarized in Table 3.

In addition to the main architecture comparison, the repository also supports exploratory changes such as reduced channel subsets and related preprocessing variations. Those experiments are treated as secondary analyses rather than the core comparison in this report, so we describe them separately in the Results section only when they contribute directly to the final model discussion.

3 Results

This section should present the main quantitative findings and point the reader to the most important comparisons. A clean structure is to start with the primary architecture comparison on the official single-user split, then follow with one or two ablations such as hyperparameter sensitivity, reduced-data training, or alternative preprocessing.

Table 4 now mixes measured values from the current training logs with ‘¿ placeholders where the corresponding metric was not yet recorded in the draft. In the final version, replace the remaining placeholders and state which model achieved the best validation CER and which model achieved the best held out test CER. If a model performs inconsistently across validation and test, discuss that explicitly rather than hiding it.

3.1 Potential split and data-efficiency studies

The course handout explicitly suggests studying how much data is needed, how many channels are needed, and how sampling rate affects performance. The table below is included so those additional experiments can be plugged into the final report without restructuring the document.

If these studies are included, the final draft should explain the split protocol precisely. For example, if you sample only a subset of train sessions, state whether the validation and test sessions remain unchanged. If the experiment changes channels or sampling rate, make clear whether the model architecture is otherwise held fixed.

3.2 Qualitative placeholders

If you later generate plots, this report is ready for:

Table 5: Sample run summary with loss, CER, and compute placeholders

Run ID	Val loss	Val CER	Test loss	Test CER	Time (hr)	Device
A1	0.71	19.8	0.93	27.4	0.9	1×T4
A2	0.65	18.3	0.86	25.6	1.1	1×T4
A3	0.58	17.6	0.81	24.9	1.0	1×T4
A4	0.60	18.1	0.83	25.1	1.4	1×T4
A5	0.56	17.9	0.80	25.0	1.7	1×T4

Table 6: Suggested future split and ablation studies

Study type	Split or subset definition	Primary metric	Purpose
Reduced-data study	25%, 50%, 75%, and 100% of train sessions	CER	Measure data efficiency
Session ablation	First 4, 8, 12, and 16 train sessions	CER	Track how added sessions improve decoding
Channel ablation	Full channel set vs. reduced channel subsets	CER	Estimate minimum useful sensor count
Sampling-rate study	Downsampled EMG features at multiple rates	CER	Measure temporal resolution sensitivity
Cross-user extension	Train on one or more extra users, test per split	CER	Evaluate generalization beyond one user

- training and validation CER over epochs,
- CER as a function of hidden size or number of layers,
- CER versus fraction of training data,
- error breakdowns showing insertion, deletion, and substitution trends.

3.3 Channel Subset Study

We next evaluated how reducing the number of EMG channels affects decoding performance. Table 7 summarizes the results summarizes the results of training the hybrid CNN+GRU model with several different channel subsets.

The full 16-channel configuration achieved the best overall performance. However, the subset containing all even-numbered channels [0,2,4,6,8,10,12,14] performed nearly as well despite using only half as many electrodes. In contrast, subsets that used only the first half [0–7] or the second half [8–15] of the channels performed worse. Additionally, the subset containing only the multiple of 4 channels [0,4,8,12] did about as well as the subsets using the only the first half or the second half of the electrodes, again despite using only half as many electrodes.

4 Discussion

The discussion should do more than restate the lowest CER. This is the section where you explain what you learned from the comparison. Questions worth answering include: Did gated recurrent models outperform the vanilla RNN? Was the improvement large enough to justify extra compute? Were some runs more stable during training? Did deeper or wider models actually help, or did they mostly overfit the single-user split?

For a strong final write-up, include at least one interpretation tied directly to the course rubric. For example, if the GRU achieves lower CER than the RNN, you might hypothesize that gating helps preserve relevant temporal context in the EMG sequence while avoiding some of the optimization difficulty of a plain recurrent model. If an LSTM does not improve over a GRU, you can discuss whether the added parameters and memory mechanisms were unnecessary for the observed time scale of the task.

Table 7: Channel subset comparison using the hybrid CNN+GRU model

Channel subset	Number of channels	Test CER
All channels	16	16.90
First half: 0–7	8	22.93
Second half: 8–15	8	25.01
Even channels: 0,2,...,12,14	8	17.55
0,4,8,12	4	23.71

Figure 1: Placeholder for a training-curve or ablation plot. Replace this box with a real figure exported from your experiment notebook or analysis script.

4.1 Sensor Redundancy and Channel Efficiency

The channel subset experiments also provide insight into how spatial information from EMG sensors affects decoding. While the model using all channels performed best, the evenly spaced subset of even-numbered channels achieved nearly identical performance despite using only half as many electrodes (17.55 CER vs. 16.90 CER). In contrast, subsets that concentrated sensors within a single contiguous region of the electrode array performed substantially worse, with the first-half [0–7] and second-half [8–15] configurations yielding 22.93 and 25.01 CER respectively (Table 7). Interestingly, a much smaller subset containing only four evenly spaced channels [0,4,8,12] achieved 23.71 CER, which is comparable to the performance of the larger contiguous 8-channel subsets. This suggests that neighboring electrodes may contain partially redundant information because EMG signals are spatially correlated across nearby sensors. As a result, maintaining spatial coverage across the forearm appears more important than simply increasing the number of adjacent electrodes. These results indicate that a reduced but well-distributed sensor configuration may retain most of the useful signal information while significantly lowering hardware complexity.

5 Future Work

We had set out to discover a model that could fit the given data for this subject significantly better than the baseline TDS model, and we found that in a TDS-RNN hybrid model. However, due time and budget constraints, there are some choices of models we considered but did not have the resources to test. These include RNN with LSTM, another type of RNN that can avoid vanishing gradients. LSTM is generally better than GRU at tracking long-term information, so in future projects, we can explore how LSTM compares with GRU.

In addition, we considered using a transformer. A transformer takes in the entire data at once and decides which parts are important using attention, compared to CNNs and RNNs that rely on processing the data sequentially. This fundamental difference means that a transformer's performance could be drastically different from anything we have compared.

Table 8: Compute and reproducibility checklist for final insertion

Item	Placeholder value	Final note to fill in
Framework	PyTorch Lightning	Version and environment file
Main device	1×GPU	Specify model, memory, and provider
Training time	~1 hr / 40 epochs	Replace with measured wall-clock time
Dataset split	Single user 89335547	Mention exact config path
Main command	`python -m emg2qwerty.train ...`	Insert exact command used

5.1 Limitations

This project currently focuses on a single user because of the compute required to train a model on the full dataset. However, that choice limits how strongly the results can generalize across users, sessions, and hardware conditions. The final discussion should also note any limits from compute budget, the number of random seeds used, and the extent of hyperparameter search. If error bars or repeated trials are not included, say so directly.

5.2 Reproducibility notes

The final report should include the exact training command, environment, and best checkpoint settings for the main comparison. One compact way to do this is to mention the training command in the discussion or appendix, for example using Hydra overrides such as `user=single_user model=gru_ctc trainer.max_epochs=40`. Add the exact command, seed, and device information once your final experiment recipe is settled.

Acknowledgments and Disclosure of Funding

This template follows the official NeurIPS 2024 style requested by the course handout and is adapted for the ECE C147/C247 final project on the emg2qwerty dataset. We would like to thank the original authors of the emg2qwerty dataset for providing the dataset and code for the initial model. We would also like to thank Professor Jonathan Kao and the ECE C147A/247A staff for providing clear course guidelines, mentorship, resources, and Google Cloud credits for our experiments.

References

[1] Alex Graves, Santiago Fernandez, Faustino Gomez, and Jurgen Schmidhuber. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376, 2006.

[2] Viswanath Sivakumar, Jeffrey Seely, Alan Du, Sean R. Bittner, Adam Berenzweig, Anuoluwapo Bolarinwa, Alexandre Gramfort, and Michael I. Mandel. *emg2qwerty: A large dataset with baselines for touch typing using surface electromyography*, 2024. arXiv:2410.20081.

[3] ECE C147/C247 staff. *ECE C147/C247, Winter 2026 project guidelines*, 2026.

A Hyperparameter tuning runs

Table 9 lists the full hyperparameter sweep used for the recurrent-model tuning study. The runs are ordered by rank, and the reported metrics are validation CER, test CER, and validation loss.

Table 9: Appendix hyperparameter tuning runs

Rank	Hidden	Layers	LR	Dropout	Val CER	Test CER	Val loss
1	128	2	0.0001	0.2	49.60	42.14	1.4942
2	128	2	0.0001	0.2	49.60	42.14	1.4942

Rank	Hidden	Layers	LR	Dropout	Val CER	Test CER	Val loss
3	128	2	0.0003	0.2	49.60	42.14	1.4942
4	256	2	0.0001	0.0	49.76	35.44	1.4594
5	256	2	0.0003	0.0	49.76	35.44	1.4594
6	256	4	0.0001	0.2	49.98	44.09	1.5561
7	128	4	0.0001	0.2	50.11	39.31	1.5446
8	128	4	0.0003	0.2	50.11	39.31	1.5446
9	128	4	0.0003	0.0	50.18	39.10	1.4771
10	128	4	0.0001	0.0	50.18	39.10	1.4771
11	128	2	0.0001	0.3	50.93	44.85	1.4902
12	128	2	0.0001	0.3	50.93	44.85	1.4902
13	128	2	0.0003	0.3	50.93	44.85	1.4902
14	256	2	0.0001	0.2	50.97	37.50	1.4991
15	256	2	0.0003	0.2	50.97	37.50	1.4991
16	256	4	0.0001	0.1	51.26	41.39	1.5438
17	256	2	0.0001	0.3	51.48	38.58	1.5244
18	256	2	0.0003	0.3	51.48	38.58	1.5244
19	128	2	0.0001	0.1	51.86	43.31	1.5267
20	128	2	0.0001	0.1	51.86	43.31	1.5267
21	128	2	0.0003	0.1	51.86	43.31	1.5267
22	128	2	0.0001	0.0	52.81	41.93	1.5367
23	128	2	0.0003	0.0	52.81	41.93	1.5367
24	128	2	0.0001	0.0	52.81	41.93	1.5367
25	128	2	0.0003	0.0	52.81	41.93	1.5367
26	256	4	0.0001	0.3	52.88	41.84	1.5574
27	128	4	0.0003	0.1	53.19	38.97	1.5226
28	128	4	0.0001	0.1	53.19	38.97	1.5226
29	128	4	0.0001	0.3	53.46	40.35	1.5911
30	128	4	0.0003	0.3	53.46	40.35	1.5911
31	256	2	0.0001	0.1	53.57	42.17	1.5326
32	256	2	0.0003	0.1	53.57	42.17	1.5326
33	256	4	0.0001	0.0	54.63	44.24	1.6220
34	256	4	0.0003	0.0	54.63	44.24	1.6220
35	128	2	0.0001	0.1	65.02	61.47	2.2227
36	128	2	0.0001	0.2	66.68	61.94	2.2677
37	128	2	0.0001	0.0	71.02	63.99	2.4464
38	128	2	0.0003	0.0	71.02	63.99	2.4464
39	128	2	0.0001	0.3	71.87	65.33	2.4487
40	?	?	?	?	100.00	100.00	7.3869
41	128	2	0.0001	0.1	104.45	78.32	4.3515
42	128	2	0.0001	0.0	107.73	79.10	4.5090
43	128	2	0.0001	0.3	107.75	80.79	4.4906
44	128	2	0.0001	0.2	108.06	80.79	4.5392

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